Science practice booklet: Grade 9 expectations

Grade 9 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 9 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Living things

Standards practised: HS-LS1-1, HS-LS1-3, HS-LS1-4, HS-LS1-5, HS-LS1-7, HS-LS3-1, HS-LS3-3

A worked example

One strand of a DNA molecule has the bases A T G G C A. Write the bases of the strand that pairs with it. Then say how many 3-base codons the original strand holds.

1. DNA bases pair in a fixed way: A pairs with T, and C pairs with G.
2. A T G G C A pairs with T A C C G T.
3. The original strand has 6 bases. Each codon is 3 bases, so 6 ÷ 3 = 2 codons.

The pairing strand is T A C C G T; the original strand holds 2 codons.

Questions

1. Name the cell part where most of the energy from food is released, and the cell part where photosynthesis happens in a plant cell.

2. Give two structures found in a plant cell but not in an animal cell, and say what each one does.

3. What is the job of the cell membrane? Why does a cell need it?

4. One DNA strand reads C G T A A G. Write the strand that pairs with it.

5. In a sample of DNA, 30 percent of the bases are adenine (A). What percentage are guanine (G)? Show your reasoning.

6. A gene is a section of DNA. What does a gene code for? If a gene region has 300 bases and every 3 bases code for one amino acid, how many amino acids are in the protein?

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7. A skin cell divides by mitosis. How many cells result, and how do their chromosomes compare with the original cell?

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8. One cell divides by mitosis, and each new cell divides again, for 4 rounds of division in total. How many cells are there at the end?

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9. Give two reasons why the cells of a young date palm and of a growing child need to divide.

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10. Write the word equation for cellular respiration, and explain how it is linked to photosynthesis.

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11. In a pea plant, tall (T) is dominant over short (t). Two tall plants, both with the genes Tt, are crossed. What fraction of their offspring are expected to be short?

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12. On a hot day in the UAE the body sweats. Explain how this helps keep the body temperature steady, and name this kind of control.

Section 2: Earth and space

Standards practised: HS-ESS2-2, HS-ESS2-4, HS-ESS2-6, HS-ESS3-1, HS-ESS3-3

A worked example

A car gives off about 120 g of carbon dioxide for every kilometre it travels. A family drives 15 000 km in a year. How many kilograms of carbon dioxide does the car give off in that year?

1. Total in grams = 120 g × 15 000 = 1 800 000 g.
2. There are 1000 g in 1 kg, so divide by 1000: 1 800 000 ÷ 1000 = 1800 kg.
3. This carbon dioxide goes into the atmosphere, where it adds to the greenhouse effect.

1800 kg (1.8 tonnes) of carbon dioxide in a year.

Questions

1. Name the four main Earth systems and give one example of each.

2. Mangrove trees grow along the coast of the UAE. Describe how they connect three Earth systems.

3. Explain how the Sun's energy drives the water cycle.

4. On a UAE beach in the afternoon, the wind usually blows from the sea towards the land. Explain why.

5. What is the difference between weather and climate? Give an example of each for the UAE.

6. Describe two ways carbon dioxide gets into the atmosphere and one way it is removed.

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7. Explain the greenhouse effect in your own words. Why is it needed for life on Earth?

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8. Why does burning more fossil fuels make Earth's average temperature rise?

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9. A home uses 600 kWh of electricity in a 30 day month. Solar panels on its roof make 5 kWh each day. What percentage of the home's electricity do the panels supply?

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10. Give one reason why solar energy is called renewable and coal is called non-renewable.

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11. The ocean takes in carbon dioxide from the air. What is one harmful result of this for coral reefs and shellfish?

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12. Ice reflects much more sunlight than dark ocean water. Explain how melting sea ice can lead to even more warming.

Section 3: Matter, forces and energy

Standards practised: HS-PS1-1, HS-PS1-2, HS-PS1-5, HS-PS1-7

A worked example

Hydrogen gas burns in oxygen to make water. The unbalanced equation is H2 + O2 → H2O. Balance it.

1. Count the atoms on each side: left has 2 H and 2 O, right has 2 H and 1 O. The oxygen is not balanced.
2. Put 2 in front of H2O: now the right side has 4 H and 2 O.
3. Put 2 in front of H2: now the left side has 4 H and 2 O. Both sides match.

2 H2 + O2 → 2 H2O

Questions

1. Name the three particles in an atom. For each, give its charge and where it is found.

2. A sodium atom has atomic number 11 and mass number 23. How many protons, neutrons and electrons does the neutral atom have?

3. Chlorine has atomic number 17. Show how its electrons are arranged in shells (2, then 8, then the rest) and say how many valence electrons it has.

4. Lithium, sodium and potassium are in the same group of the periodic table. Why do they react in similar ways?

5. Helium and neon are very unreactive gases. Use their outer electrons to explain why.

6. Describe how sodium and chlorine atoms form the ionic compound sodium chloride (table salt).

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7. In a water molecule (H_2O), the atoms share electrons. What is this kind of bond called, and how is it different from an ionic bond?

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8. Balance this equation for the burning of magnesium: $Mg + O_2 \rightarrow MgO$

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9. Balance the equation for burning methane (natural gas): $CH_4 + O_2 \rightarrow CO_2 + H_2O$

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10. 24 g of magnesium burns completely with 16 g of oxygen. What mass of magnesium oxide is made, and which law does this show?

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11. An iron nail left outside near the coast turns into rust, and the rust weighs more than the nail did. Has mass been created? Explain.

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12. Photosynthesis is written as $6\ CO_2 + 6\ H_2O \rightarrow C_6H_{12}O_6 + 6\ O_2$. Show that this equation is balanced by counting the atoms of each element on each side, and give one reason why changing the temperature or the amount of a reactant can change how fast a reaction goes.

Answer guide for parents

Section 1: Living things

1. Mitochondria release energy (cellular respiration). Chloroplasts carry out photosynthesis.
2. Any two of: cell wall (supports and protects the cell), chloroplasts (make food by photosynthesis), a large central vacuole (stores water and keeps the cell firm).
3. It controls which substances enter and leave the cell. The cell needs it to take in nutrients and oxygen, remove wastes and keep its inside different from the outside.
4. G C A T T C (C pairs with G, and A pairs with T).
5. 20 percent. Thymine (T) is also 30 percent because A pairs with T, so A + T = 60 percent. The other 40 percent is C and G in equal amounts, so G = 20 percent.
6. A gene codes for a protein (a chain of amino acids). 300 ÷ 3 = 100 amino acids.
7. Two cells result. Each has the same number and kind of chromosomes as the original cell, so they are genetically identical to it.
8. 2 × 2 × 2 × 2 = 16 cells (the number doubles each round).
9. For growth (more cells make the body larger) and to repair or replace damaged and worn-out cells.
10. Glucose + oxygen makes carbon dioxide + water + energy released. Photosynthesis makes the glucose and oxygen that respiration uses, and respiration makes the carbon dioxide and water that photosynthesis uses.
11. One quarter (25 percent). The offspring are TT, Tt, Tt and tt in a 1 : 2 : 1 pattern, and only tt is short.
12. Sweat evaporates from the skin and takes heat away, so the body cools and stays near 37 degrees Celsius. This is homeostasis, kept steady by negative feedback: a rise in temperature causes a response that lowers it.

Section 2: Earth and space

1. Geosphere (rocks, the Hajar mountains), hydrosphere (water, the sea), atmosphere (air, clouds) and biosphere (living things, mangroves).
2. They take in carbon dioxide from the atmosphere and water from the sea (hydrosphere), build it into their living tissue (biosphere), and their roots hold the muddy soil in place (geosphere).
3. The Sun heats sea and land water so it evaporates into water vapour. The vapour rises and cools, condenses into clouds, and falls as rain or other precipitation, then flows back to the sea.
4. Land heats up faster than water, so the air over land warms and rises. Cooler, denser air from over the sea moves in to take its place, which we feel as a sea breeze.
5. Weather is the condition of the air at a certain place and time (for example, a foggy morning today). Climate is the pattern of weather over many years (for example, hot, dry summers).
6. Added by: burning fossil fuels, and respiration by living things (also decay and volcanoes). Removed by: photosynthesis by plants and algae (or by dissolving in the ocean).

7. Gases such as carbon dioxide and methane absorb some of the heat Earth gives off and send part of it back, which keeps the surface warm. Without this natural effect Earth would be too cold for most life.
8. Burning releases extra carbon dioxide, so more heat is trapped in the atmosphere and the average temperature goes up over time.
9. 5 × 30 = 150 kWh. 150 ÷ 600 = 0.25, so 25 percent.
10. Sunlight keeps arriving every day and cannot be used up, but coal took millions of years to form and cannot be replaced at the rate we use it.
11. Dissolved carbon dioxide makes the seawater more acidic (ocean acidification). This makes it harder for corals and shellfish to build their hard shells and skeletons.
12. As ice melts, more dark water is uncovered and it absorbs more sunlight instead of reflecting it. The extra heat melts more ice. This is a positive feedback loop that increases the change.

Section 3: Matter, forces and energy

1. Protons: positive, in the nucleus. Neutrons: no charge, in the nucleus. Electrons: negative, moving around the nucleus.
2. Protons 11, electrons 11 (a neutral atom), neutrons 23 − 11 = 12.
3. 2, 8, 7. It has 7 valence electrons (the electrons in the outer shell).
4. Atoms in the same group have the same number of valence electrons (one each here), and it is the outer electrons that decide how an atom reacts.
5. Their outer shells are full, which is a stable arrangement, so their atoms do not need to gain, lose or share electrons.
6. A sodium atom gives one electron to a chlorine atom. Sodium becomes a positive ion (Na^+) and chlorine becomes a negative ion (Cl^-). The opposite charges attract and hold the ions together.
7. A covalent bond. In a covalent bond atoms share electrons (usually between non-metal atoms). In an ionic bond electrons are transferred from one atom to another.
8. $2\ Mg + O_2 \rightarrow 2\ MgO$ (2 Mg and 2 O on each side).
9. $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ (1 C, 4 H and 4 O on each side).
10. 24 + 16 = 40 g. This shows the law of conservation of mass: the total mass of the reactants equals the total mass of the products.
11. No. Iron has joined with oxygen (and water) from the air to make rust, so the extra mass came from the oxygen. Atoms are rearranged, not created or destroyed.
12. Left: C 6, H 12, O 12 + 6 = 18. Right: C 6, H 12, O 6 + 12 = 18. They match, so it is balanced. A higher temperature or more reactant means particles collide more often and with more energy, so the reaction goes faster.